6 June 1955
SUPERSEDING
NAVORD OS 3638 B
27 JANUARY 1954

MILITARY SPECIFICATION

INITIATOR, DETONATING CORD, MARK 2 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of Mark 2 Mod 0 Detonating Cord Initiator and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-L-73	-- Lacquer-Enamel, Lusterless,
MIL-A-101	-- Adhesive, Water-Resistant, for Sealing Fiberboard Boxes.
JAN-P-106	-- Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
JAN-P-108	-- Packaging and Packing for Overseas Shipment - Boxes; Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN-C-388	-- Compound, Sealing N.R.C.
MIL-V-16399	-- Varnish, Moistureproof

STANDARDS

MIL-STD-105	-- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-- Marking for Shipment and Storage.

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DRAWINGS

BUREAU OF ORDNANCE

LD SK No. 165321 -- Initiator (Detonating Cord) Mk 2 Mod 0,
DWG 447888 -- Initiator (Detonating Cord) Mk 2 Packaging,
DWG 448912 -- Tetryl Head,
And all documents listed on the above LD and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950)".)

3. REQUIREMENTS

3.1 General.

3.1.1 The Detonating Cord Initiator covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD No. 165321 and all documents listed thereon.

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3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Cleanliness.- All component inert parts shall be thoroughly cleaned before assembly. Care shall be taken to prevent the inclusion of dirt or other foreign matter in the assembly. No application of cement, varnish, or luting shall be made by the inert parts manufacture.

3.3 Zinc die casting.- When zinc die castings are used for the detonator container, cap holder or tetryl head, the dies and techniques of production should be such that no surface, external or internal, shall show a defect due to lack of metal and that internal porosity shall not cause a reduction in weight exceeding one-half of one percent of the maximum possible for the piece. All gates, fins and vents shall be removed in such a way as to cause no injurious stress in the piece and no permanent deformation discoverable by careful measurement. The remaining portions of gates, fins and vents shall not extend beyond the normal surface more than 0.005 inch and shall be treated as to leave no edges or points liable to cause personal injury in handling. The detonator container shall be entirely free from protrusions into any portion of the through hole. The tetryl head shall have no web, fin, or sharp edge extending into or across the notch. If produced by die casting, any flash or sharpness of the ends of the notch shall be carefully removed.

3.4 Tetryl head.- X-ray examination of the tetryl head shall show no defects in the thin web between the notch and the tetryl space. Testing and acceptance shall be in accordance with 4.5.1

3.5 Explosive parts.

3.5.1 Detonator.- The sensitive end of the detonator which is closed by the detonator disc and detonator sealing disc shall be given a thin coat of red N.R.C. compound conforming to the requirements of Specification JAN-C-388.

3.5.2 Assembly of detonator into detonator container.- The exterior surface of the detonator shall be coated with moistureproof varnish conforming to the requirements of Specification MIL-V-16399 before inserting into the detonator container. After insertion, excess varnish shall be removed and the remainder allowed to dry thoroughly.

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3.6 Painting.- After loading, the outside of each initiator shall be given one (1) coat of lusterless olive drab lacquer enamel conforming to the requirements of Specification JAN-L-73, Grade 1.

3.7 Functioning.- The initiator shall function after being submerged in ten feet of water for one hour. Testing and acceptance shall be in accordance with 4.5.2.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling.- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot.- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample.- Before entering into quantity production, a preproduction sample of twenty-five (25) Complete Mark 2 Mod 0 Detonating Cord Initiator shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These initiators shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract.- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot.- The standard production lot shall consist of five hundred (500) initiators plus such excess as is required for sampling and tests. The production lot shall be used for delivery and identification of initiators and is not related to the inspection lot defined in Standard MIL-STD-105.

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4.4 Inspection.- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Tests.

4.5.1 X-ray examination of tetryl heads.- During production of the first one thousand (1,000) tetryl heads, Bureau of Ordnance Drawing No. 448912, at least one hundred (100) pieces shall be examined by X-ray to reveal any tendency to a defect in the thin web between the notch and the tetryl space. Thereafter, at the discretion of the Inspector, this examination may be required; especially on starting a new melt of metal, or on starting up after interrupted operation, some of the first product proposed for acceptance shall be so examined.

4.5.2 Functioning test.- Ten completed initiators, selected at random from each lot shall be tested as specified in 4.5.2.1. Should one of these ten initiators fail to cause complete detonation of the cord, an additional ten initiators from the same lot shall be tested. More than one failure out of 20 tested shall cause rejection of the entire lot, from which the samples were taken.

4.5.2.1 Test procedure.- Immerse the initiators in ten feet of water for one hour. After one hour withdraw the initiators from the water and remove the cap holders. Insert a detonating cord in the slot of the tetryl head and then screw into the detonator holder any standard mechanical firing device with coupling base attached and fire.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Ten Mark 2 Mod 0 Detonating Cord Initiators shall be packaged snugly and hermetically sealed in a labeled and marked metal can as shown on Bureau of Ordnance Drawing No. 447888. Ten sealed cans shall be packaged in a two by one by five arrangement, in a corrugated or solid fiberboard carton conforming to the requirements of Specification JAN-P-108, Style RSC, Type W6c or W6s, except that metal strapping will not be required. Carton shall be designed so that the contents fit

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snugly and shall be approximately 6 by 5 1/4 by 15 inches, inside dimensions. Closure of cartons shall be effected by application of water resistant adhesive, conforming to the requirements of Specification MIL-A-101, applied over the entire area of contact between the flaps.

5.2 Packing.- Five cartons, as specified in 5.1 shall be packed in a nailed wooden box conforming to the requirements of Specification JAN-P-106, Style 4. Boxes shall be designed so that the contents fit snugly and shall be approximately 6 1/4 by 27 by 15 1/4 inches, inside dimensions.

5.3 Marking.

5.3.1 The cartons specified in 5.1 shall be stamped or stenciled in 1/2 inch high letters as follows:

In black ink;
"100 INITIATORS, DETONATING CORD, MARK 2 MOD 0"
In red ink;
"INITIATORS (BLASTING CAPS)
HANDLE CAREFULLY
DO NOT STORE OR LOAD WITH
ANY HIGH EXPLOSIVE"

5.3.2 The boxes specified in 5.2 shall be stenciled in 1/2 inch high letters as follows:

In black;
"500 INITIATORS, DETONATING CORD, MARK 2 MOD 0"
In red;
"INITIATORS (BLASTING CAPS)
HANDLE CAREFULLY
DO NOT STORE OR LOAD WITH
ANY HIGH EXPLOSIVE"

5.3.3 In addition to any special markings required by the contract or order, and that specified in 5.3.1 and 5.3.2, each container shall be marked in accordance with the requirements of Code of Federal Regulations, 49 CFR 71-78, and Standard MIL-STD-129.

6. NOTES. -- There are none applicable.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the

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United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 1 APR 63 *REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE* S-0270

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